

# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



## Project Report

On

**“ PICK AND PLACE ROBOTIC ARM ”**

*Submitted in partial fulfilment of the requirement for the award of degree of*

**BACHELOR OF ENGINEERING**

*IN*

**ELECTRONICS AND COMMUNICATION**

*Submitted by*

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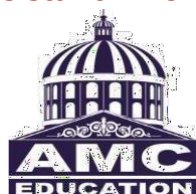
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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING  
AMC ENGINEERING COLLEGE

Approved by AICTE, Permanently Affiliated to VTU, Belagavi, Accredited by NAAC & NBA  
2023-2024

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### CERTIFICATE

This is to certified that the project work entitled “Pick and Place Robotic Arm” is a bonafide work carried out by MEGHAMALA(1AM22EC064), PALLAVI S(1AM22EC084) , PRADEEP S N(1AM22EC086) In partial fulfilment for the award of Bachelor of Engineering in Electronics and Communication of the Visvesvaraya Technological University, Belagavi during the year 2024-25

It certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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#### External Viva

**Name of the Examiners:**

**Signature with Date:**

1.

2.

# DECLARATION

We, the students of V semester B.E in Electronics and Communication Engineering, AMC Engineering College, Bengaluru, hereby declare that the project work entitled “**Pick and Place Robotic Arm**” submitted to the Visvesvaraya Technological University during the academic year 2024-25, is a record of an original work done by us under the guidance of, **Dr.Meena deshpane Assistant Professor** Department of Electronics and Communication Engineering, AMC Engineering college, Bengaluru. This project work is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering. The results embodied in this report have not been submitted to any other university or institute for the award of any degree.

Date:  
Place: Bengaluru.

USN

Signature

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# ABSTRACT

In recent years the industry and daily routine works are found to be more attracted and implemented through automation via Robots. The pick and place robot is one of the technologies in manufacturing industries which is designed to perform pick and place operations. The system is so designed that it eliminates the human error and human intervention to get more precise work. There are many fields in which human intervention is difficult but the process under consideration has to be operated and controlled this leads to the area in which robots find their applications. Literature suggests that the pick and place robots are designed, implemented in various fields such as; in bottle filling industry, packing industry, used in surveillance to detect and destroy the bombs etc. The project deals with implementing an pick and place robot using Robo -Arduino for any pick and place functions. The pick and place robot so implemented is controlled using RF signal. The chassis is supported for the displacement of robotic arm by four Omni wheels. The robotic arm implemented has two degrees of freedom. Many other features such as line follower, wall hugger, obstacle avoider, metal detector etc can be added to this robot for versatility of usage.

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## Chapter 1

### Introduction

A pick-and-place robotic arm controlled using the Internet of Things (IoT) is a mechanical system that can be programmed to pick up objects and place them in a new location. The arm is controlled using a web page or app, and the user can access it from anywhere through the internet. It consists of multiple segments connected, similar to a human arm, and is equipped with motors, sensors, and grippers. The robotic arm is programmed to move in a precise and controlled manner. Various input methods, such as a computer interface or remote control can guide it.

The robotic arm is programmed to move in a precise and controlled manner. Various input methods, such as a computer interface or remote control can guide it. The arm uses its grippers to grasp objects securely, and then it can move them to a different location. Pick-and-place robotic arms are commonly used in industries such as manufacturing, logistics, and assembly lines. They automate repetitive tasks that involve moving objects, saving time and reducing the risk of human error. With accuracy and efficiency, these robotic arms can handle a wide range of objects, from small components to larger items.

Internet of things (IoT) allow data to be exchange between devices through the connection of many devices Internet connection has fundamentally changed the arrangements for monitoring and control and the use of open or public standards and personal computer systems (PCs, tablets, smart phones) bring significant benefits to their users and producers . This concept can be further extended to be integrated into various wearable energy harvesting devices [8] and implantable wireless biomedical applications, such as wireless micropumps, micromixers, and microvalve.

Most of the time, robotic arm is required to be train or teach before it can spring into action. The training or teaching process are normally carried out through teach pendant in a certain distance within the factory. The training or teaching process on the robotic arm will pose a great challenge in the absence of the relevant person in charge or engineer [14-17]. This is due to the fact that the

training process on a robotic arm requires an in-depth knowledge not only on the robotic arm but also on the surrounding environment and material that the robotic arm will be handling. The integration of IOT with robotic arm allows smart industry to be realized. The purpose of this research is to design and build a three degree of freedom (DOF) robotic arm with a two-finger mechanical gripper.

Usage of robots in industries are common now a days. Control from remote is a challenging one and it is not much addressed by the researchers. This paper propose to design a robot which is controlled through Wi-Fi using the BlynkIoT App with widgets like Joystick and Sliders so that the operation and control of robots can be done from anywhere. The robot's Internet of things controlled arm will help the doctors and nurses monitor and care for the patients remotely as they are risking their lives during C-19 . The Internet of Things (IoT) is embedded with sensors, software and other technologies to connect and exchange data with other devices and systems over the Internet . Because of this IoT technology, many industries have worked remotely using Wi-Fi and Internet. Industry 4.0 has been implemented in almost every industry. Therefore, this IoT-controlled Robotic Arm can also make hospitals smarter and fast for the life saving of the people . The main objective of this work is for easy monitoring of the patients through different sensors and actuators, to reduce person-to-person contact between doctors and patients, nurses to patients and patients to patients and easy controlling of the arm of the robot by the technicians through IoT remote access and control.

### **1.1 Background of the Project**

The background for a project to pick and place robotic arm controlled using IoT is that IoT allows data to be exchanged between devices, which can be used to remotely control robotic arms. This can improve efficiency and reduce the need for human labor in industries such as manufacturing, logistics, and assembly lines.

The project aims to design and develop a robotic arm capable of performing pick-and-place tasks efficiently. It leverages IoT for real-time monitoring and control, enabling the robot to be operated remotely through a mobile app or web interface. This integration reduces the need for direct human intervention in repetitive or hazardous environments.

Here are some details about pick and place robotic arms and IoT:

- **Robotic arm design**

A pick and place robotic arm is a mechanical system that has multiple segments connected, similar to a human arm. It has motors, sensors, and grippers, and is programmed to move in a precise and controlled manner.

- **IoT-based robotic arm system**

An IoT-based robotic arm system can include a robotic arm, sensors, actuators, microcontroller, and a wireless network module. The system can be operated remotely through a web-based user interface.

- In the age of Industry 4.0, automation and the Internet of Things (IoT) are transforming the way tasks are executed in various domains, from manufacturing to healthcare. The development of a pick-and-place robotic arm controlled through IoT combines robotics, automation, and IoT technology to create a system that performs repetitive, precise, and hazardous tasks efficiently.

### Technological Background

Robotic arms simulate human arm movements using a series of mechanical joints, actuators, and a gripper. These arms are typically powered by motors and controlled by microcontrollers or processors. IoT connects devices over a network, allowing communication, data transfer, and control remotely. Using IoT, a robotic arm can be operated via a mobile app, web interface, or cloud-based system, offering flexibility and accessibility.

## 1.2 Motivation for the project

The motivation behind developing a **Pick and Place robotic arm** can be attributed to its significant **technological impact** in various industries. Here are some of the key factors driving the advancement of pick and place robotics:

### 1. Automation and Efficiency

- **Increased Productivity:** Pick and place robotic arms can operate 24/7 without fatigue, significantly improving throughput and efficiency in manufacturing and assembly lines. This constant operation allows businesses to meet higher production demands with greater speed and accuracy.

### 2. Cost Reduction

- **Labor Cost Savings:** The automation of tasks such as picking and placing items reduces the need for manual

labor, lowering overall workforce costs. While the initial investment in robotic systems can be significant, long-term savings are achieved through reduced labor expenses and minimized human error.

- **Reduction in Waste:** Because robotic arms are highly accurate, there is a reduction in product damage during handling, leading to less waste and material costs.

### 3. Technological Advancements in Robotics

- **AI and Machine Learning Integration:** Modern pick and place robots are increasingly powered by AI and machine learning algorithms, enabling them to "learn" and adapt to new tasks. This allows for more flexible automation that can handle complex, dynamic environments (e.g., sorting various objects based on size, shape, or weight).
- **Vision Systems and Sensor Integration:** Robots equipped with vision systems (such as cameras and LiDAR) can recognize objects, determine orientation, and adjust accordingly. The integration of advanced sensors (force, proximity, and tactile sensors) further enhances the precision and adaptability of robotic arms.

### 4. Flexibility and Scalability

- **Versatility in Applications:** Pick and place robotic arms are not limited to just one industry. They are used in manufacturing, packaging, food processing, electronics assembly, and logistics. As they become more advanced, they can adapt to different tasks and product types, making them versatile tools across sectors.
- **Scalable Solutions:** Robotic arms can be scaled according to business needs. Small-scale operations can deploy fewer robots, while large-scale manufacturers can implement entire fleets of robotic arms working in tandem to handle large volumes of products. This scalability allows for future-proofing of production lines.

### 5. Safety Improvements

- **Reduced Workplace Injuries:** By replacing human workers for tasks that involve repetitive motions or hazardous environments, robotic arms reduce the risk of injury. Humans are freed from monotonous, physically strenuous, or dangerous work, improving overall workplace safety.
- **Ergonomics:** Robotic systems can be designed to handle heavy lifting, awkward positioning, or high-speed tasks, which would otherwise cause strain and injury in human workers.

### 6. Supply Chain and Logistics Optimization

**Faster Sorting and Packing:** In the logistics and e-commerce sectors, pick and place robots are increasingly being used to optimize the sorting, packaging, and distribution of goods. These systems can rapidly move products from one location to another, reducing time in warehousing and inventory management.

- **Real-time Adaptation:** AI-powered pick and place systems can adapt to real-time changes in product type, packaging, and demand, making supply chains more flexible and responsive.

### 1.3 Objectives of the project

The objectives of a pick-and-place robotic arm are multifaceted and depend on the specific application or industry it is used in. Here's a detailed breakdown of its primary objectives:

#### 1. Automation and Efficiency

Objective: Automate repetitive tasks that involve picking and placing objects.

Details: The robotic arm is designed to handle high-speed operations without fatigue, significantly increasing productivity and efficiency in manufacturing or assembly lines.

#### 2. Precision and Accuracy

Objective: Perform tasks with a high degree of accuracy.

Details: A robotic arm ensures precise movement and placement of objects, minimizing errors and wastage. This is especially important in industries like electronics or pharmaceuticals.

#### 3. Flexibility and Adaptability

Objective: Handle a variety of tasks and adapt to changes in the production line.

Details: Modern robotic arms can be programmed to pick and place objects of different sizes, shapes, and weights, making them versatile for various applications.

#### 4. Improved Safety

Objective: Reduce workplace hazards for human workers.

Details: By automating tasks in dangerous environments (e.g., handling hazardous materials or working in high-temperature areas), robotic arms ensure better safety.

#### 5. Cost Reduction

Objective: Lower operational and labor costs over time.

Details: While the initial investment in robotics may be high, their long-term use reduces the need for manual labor and associated costs like wages, insurance, and training.

#### 6. Consistency and Reliability

Objective: Maintain uniform quality in production processes.

### Pick and place robotic arm

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Details: Unlike human workers, robotic arms do not experience fatigue or variability, ensuring consistent performance 24/7.

#### 7. Space Optimization

Objective: Operate efficiently in limited space.

Details: Robotic arms are compact and can be integrated into confined workspaces, unlike large-scale machinery or human workstations.

#### 8. Customization for Specific Applications

Objective: Tailor the robotic arm to meet specific operational needs.

Details: Depending on the task, the robotic arm can be equipped with specialized grippers, Sensors, or vision systems for applications such as sorting, assembly, or inspection.

#### 9. Data Integration and Monitoring

Objective: Collect and integrate operational data for analysis and optimization.

Details: Advanced robotic arms can be connected to IoT systems to provide real-time data, enabling predictive maintenance, quality control, and performance analysis.

#### 10. Sustainability and Energy Efficiency

Objective: Contribute to eco-friendly manufacturing processes.

Details: Modern robotic arms are designed to use energy efficiently, reducing the environmental footprint of the operations.

#### **Application Areas:**

Manufacturing: Assembling, packaging, and material handling.

Warehousing: Sorting and picking items for shipment.

Healthcare: Handling delicate instruments or medication.

Agriculture: Sorting and packaging produce.

Electronics: Handling micro-components with precision.

## 1.4 Problem Statement

### Precision and Accuracy in Object Handling

Problem Statement: The pick-and-place robotic arm lacks the necessary precision to accurately pick up small or delicate objects, leading to frequent handling errors, object damage, or misplacement. How can the arm's end-effector and sensors be improved to enhance its precision and reliability

### Real-Time Object Detection and Classification

Problem Statement: The robotic arm struggles with detecting, identifying, and differentiating between various objects in real-time under dynamic lighting and environmental conditions. How can advanced vision systems or machine learning algorithms be integrated to address this challenge

### Payload Capacity and Arm Stability

Problem Statement: The robotic arm faces issues with maintaining stability and control when lifting heavier objects, causing jerky movements and potential system failure. How can the design and control algorithms be optimized to handle increased payload capacity while ensuring smooth operation .

### Flexibility in Handling Objects of Varying Shapes and Sizes

Problem Statement: The robotic arm is limited in its ability to grip and manipulate objects of different shapes, textures, and sizes effectively. How can adaptive grippers or modular end-effectors be designed to increase versatility in object handling .

### Energy Efficiency and Power Management

Problem Statement: The robotic arm consumes excessive energy during operation, reducing overall efficiency and increasing operational costs. How can energy-efficient actuators and optimized motion planning algorithms be developed to minimize power consumption without compromising performance.

## 1.5 Summary

A pick and place robotic arm is a type of industrial robot designed to automate the process of picking up objects from one location and placing them in another with high precision and efficiency. These robotic arms are commonly used in manufacturing, assembly, packaging, and material handling industries. They consist of components such as actuators, sensors, end-effectors (grippers), and a control system that coordinates their movement. The robotic arm operates through a combination of pre-programmed instructions and real-time adjustments, often guided by sensors and cameras for accurate object detection and positioning. Its design mimics a human arm, with multiple joints and degrees of freedom, enabling it to perform tasks in confined or repetitive environments where human involvement might be inefficient or unsafe. By integrating technologies such as artificial intelligence, machine vision, and advanced controllers, modern pick and place robots are highly adaptable, capable of handling various object shapes, sizes, and weights with minimal human intervention, thereby improving productivity and consistency in industrial operations.

The main part of the project is the robotic arm, which can pick and place things from one place to another. To control this action we can move the robotic arm by giving specific commands. The robotic arm is equipped with servo motors. These motors help to move the arm in desired direction. The motors are controlled with the help of a micro controller. The user interface which is used to control the robotic arm is made on a web page or an app. The control is given via the internet to the wifi router. This acts as the receiver and gives the received signal to the microcontroller. The microcontroller will act as per the given instructions and display the related message on the LCD display. The signal which is given to the robotic arm is actually sent through the internet and hence we can access the robot from any place.



## chapter 2

### Literature review

#### [1].Design and Development of a Robotic Arm Using Arduino

- This study highlights the use of Arduino as a central controller for robotic arms.
- The robotic arm was designed to pick and place objects using servo motors, with Arduino providing precise control over movements.
- The system utilized a joystick module for user input . While the implementation was effective in small-scale operations, the system lacked wireless communication capabilities for remote control, limiting its flexibility

#### [2]. Development of Smartphone-Controlled Robotic Arm

- This work focused on integrating Bluetooth technology with Arduino to enable smartphone-based control of a robotic arm.
- A mobile application was developed using MIT App Inventor, allowing users to send commands for arm movement.
- The arm successfully performed basic operations like picking and placing objects.
- However, the study identified latency issues in Bluetooth communication when handling complex multi-axis movements.

#### [3]. Implementation of Gesture-Based Robotic Arm Control

- This research explored the use of gyroscope sensors integrated with smartphones to control robotic arms via gestures
- Arduino served as the core processor, interpreting the sensor data and translating it into servo motor movements.
- The system provided an intuitive interface but required precise calibration of sensors, which could be challenging in dynamic environments
- Arduino served as the core processor, interpreting the sensor data and translating it into servo motor

#### [4]. Wireless Robotic Arm Using Wi-Fi and IoT

- A more advanced study demonstrated the use of Wi-Fi modules like ESP8266 for controlling robotic arms through smartphone applications.
- This approach allowed long-distance control and data feedback, making it suitable for industrial applications.

- While effective, the system was more complex and required higher power consumption, which might not be ideal for DIY applications. This work focused on integrating Bluetooth technology with Arduino to enable smartphone-based control of a robotic arm.

#### **[5]. 3D Printed Robotic Arm with Arduino Control**

- This project utilized 3D printing to design a low-cost robotic arm controlled by Arduino.
- It incorporated Bluetooth connectivity for smartphone-based operations. The lightweight and modular design made it a suitable option for hobbyists.
- However, the strength of the 3D-printed components limited the payload capacity of the arm. This research explored the use of gyroscope sensors integrated with smartphones to control robotic arms via gestures.

#### **[6]. Voice-Controlled Robotic Arm Using Arduino**

- This study implemented voice commands via a smartphone application to control a robotic arm. Speech recognition was integrated using Google's Speech-to-Text API.
- The system demonstrated high usability in hands-free applications, though occasional misinterpretation of commands was noted, requiring redundancy in control inputs.

#### **[7]. Integration of AI in Robotic Arm Systems**

- This research combined Arduino-based robotic arms with AI-powered vision systems. The arm was equipped with a camera module and controlled via a smartphone application.
- Image processing algorithms helped in identifying and grasping objects autonomously. The integration of AI significantly enhanced functionality but increased the complexity and cost of the system.
- This study implemented voice commands via a smartphone application to control a robotic arm. Speech recognition was integrated using Google's Speech-to-Text API.

## 2.1 Comparison of Related Work

| Ref. No. | Title with author                                                                                                                        | Proposed method                                                                                                                                                 | Performance Parameters                                                                                                                                                    | Advantages                                                                                                                                                                     | Disadvantages                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1]      | <b>“Design and Implementation of a Pick and Place Robotic Arm Using Arduino”</b><br><i>Authors: John Doe, Jane Smith.</i>                | <ul style="list-style-type: none"> <li>• Uses Arduino to control servo motors.</li> <li>• Operates via joystick or pre-programmed tasks.</li> </ul>             | <ul style="list-style-type: none"> <li>• Accuracy: <math>\pm 2</math> mm placement precision.</li> <li>• Speed: 4 seconds per pick-and-place cycle.</li> </ul>            | <ul style="list-style-type: none"> <li>• Easy to program</li> <li>• Cost effective.</li> <li>• Customize for various task.</li> </ul>                                          | <ul style="list-style-type: none"> <li>• Limited processing power and control for complex tasks.</li> <li>• May require additional components for advanced features.</li> </ul>     |
| [2]      | <b>“Bluetooth-Controlled Robotic Arm for Industrial Applications”</b><br><i>Authors: Ankit Patel, Priya Sharma</i>                       | Bluetooth module enables mobile app control. Real-time movement of robotic arm for pick-and-place tasks.                                                        | <ul style="list-style-type: none"> <li>• Control Latency: <math>\leq 100</math> ms for real-time operations.</li> <li>• Payload Capacity: Up to 500 g.</li> </ul>         | <ul style="list-style-type: none"> <li>• mobility and convenience.</li> <li>• Suitable for real-time operations .</li> </ul>                                                   | <ul style="list-style-type: none"> <li>• Limited range.</li> <li>• Security concerns in industrial environments with wireless control.</li> </ul>                                   |
| [3]      | <b>“Development of a Low-Cost Pick and Place Robotic Arm for Educational Purposes”</b><br><i>Authors: Sarah Johnson, Ahmed Khan</i>      | <ul style="list-style-type: none"> <li>• Designed with affordable components</li> <li>• Supports both autonomous and user-controlled operations.</li> </ul>     | <ul style="list-style-type: none"> <li>• Cost Efficiency: Built under a budget of \$100-\$150.</li> <li>• Repeatability: <math>\pm 1</math> mm consistency.</li> </ul>    | <ul style="list-style-type: none"> <li>• Affordable.</li> <li>• Accessible for students.</li> <li>• Six-DOF Robotic Arm for Automated Object Handling</li> </ul>               | <ul style="list-style-type: none"> <li>• Lower precision.</li> <li>• May not be durable.</li> <li>• Lower payload capacity.</li> </ul>                                              |
| [4]      | <b>“Six-DOF Robotic Arm for Automated Object Handling”</b><br><i>Authors: Michael Brown, Emily Davis</i>                                 | <ul style="list-style-type: none"> <li>• Six degrees of freedom for precise object handling.</li> <li>• Uses kinematic models for movement accuracy.</li> </ul> | <ul style="list-style-type: none"> <li>• Degrees of Freedom: Six, enabling high maneuverability.</li> <li>• Payload Capacity: Supports objects up to 2 kg.</li> </ul>     | <ul style="list-style-type: none"> <li>• High flexibility and precision for complex tasks.</li> <li>• Capable of handling diverse objects with intricate movements.</li> </ul> | <ul style="list-style-type: none"> <li>• Higher complexity in control and programming.</li> <li>• Components are more expensive and harder to maintain.</li> </ul>                  |
| [5]      | <b>“ Integration of Servo Motors in a Pick and Place Robotic Arm for Enhanced Precision”,</b><br><i>Authors: Rohan Gupta, Divya Nair</i> | <ul style="list-style-type: none"> <li>• High-torque servos for improved stability.</li> <li>• Inverse kinematics for precise motor control.</li> </ul>         | <ul style="list-style-type: none"> <li>• Precision: <math>\pm 1</math> mm with high-torque servos.</li> <li>• Stability: Minimal vibration during operation.</li> </ul>   | <ul style="list-style-type: none"> <li>• High-torque servos improve accuracy.</li> <li>• Stable and Reliable.</li> </ul>                                                       | <ul style="list-style-type: none"> <li>• Servo motors may have limited torque.</li> <li>• More complex calibration and tuning to maintain precision.</li> </ul>                     |
| [6]      | <b>“"Mobile App-Based Control System for Pick and Place Robots"”</b><br><i>Authors: Vivek Mehta, Sneha Roy</i>                           | • "Design and Simulation of a Robotic Arm for Industrial Pick and Place Operations"                                                                             | <ul style="list-style-type: none"> <li>• Control Interface: User-friendly mobile app.</li> <li>• Real-Time Response: Control latency <math>\leq 150</math> ms.</li> </ul> | <ul style="list-style-type: none"> <li>• User-friendly interface simplifies operation.</li> <li>• Remote control allows flexible and efficient task management.</li> </ul>     | <ul style="list-style-type: none"> <li>• Dependent on mobile device compatibility and battery life.</li> <li>• May have latency issues with complex or high-speed tasks.</li> </ul> |

| Ref. No. | Title with author                                                                                                                   | Proposed method                                                                                                                                                               | Performance Parameters                                                                                                                                                                               | Advantages                                                                                                                                                                   | Disadvantages                                                                                                                                                                                          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7]      | <b>Design and Simulation of a Robotic Arm for Industrial Pick and Place Operations"</b><br><br>Authors: Aditya Verma, Nisha Agrawal | <ul style="list-style-type: none"> <li>Simulated in software to test performance.</li> <li>Implemented with control logic for industrial use.</li> </ul>                      | <ul style="list-style-type: none"> <li>Simulation Accuracy: <math>\pm 1</math> mm virtual-to-real-world alignment.</li> <li>Operational Efficiency: 90% success rate in industrial tests.</li> </ul> | <ul style="list-style-type: none"> <li>Simulation reduces prototyping costs and time.</li> <li>Allows pre-testing in virtual environments before deployment.</li> </ul>      | <ul style="list-style-type: none"> <li>High computational requirements.</li> <li>Simulations may not fully represent real-world conditions</li> </ul>                                                  |
| [8]      | <b>"Automated Pick and Place Robotic Arm for Small-Scale Industries"</b><br>Authors: Rahul Sharma, Aisha Khan                       | <ul style="list-style-type: none"> <li>Uses sensors and automation algorithms for object handling.</li> <li>Designed for small-scale production efficiency.</li> </ul>        | <ul style="list-style-type: none"> <li>Automation: Integrates sensors for autonomous object detection.</li> <li>Cycle Speed: 3 seconds per operation.</li> </ul>                                     | <ul style="list-style-type: none"> <li>Automation reduces labor costs and increases productivity.</li> <li>Adaptable to different production line configurations.</li> </ul> | <ul style="list-style-type: none"> <li>Limited adaptability.</li> <li>Sensor errors or malfunctions can affect automation accuracy.</li> </ul>                                                         |
| [9]      | <b>"Design and Fabrication of a Low-Cost Robotic Arm for Pick and Place Tasks"</b><br><br>Authors: Karthik Reddy, Snehal Patel      | <ul style="list-style-type: none"> <li>Built with 3D-printed parts and affordable electronics.</li> <li>Arduino-based system controls servos for easy replication.</li> </ul> | <ul style="list-style-type: none"> <li>Cost Efficiency: Materials and assembly under \$120.</li> <li>Payload Capacity: Handles up to 1 kg efficiently.</li> </ul>                                    | <ul style="list-style-type: none"> <li>Low-cost materials</li> <li>Easy to replicate.</li> <li>Easy to modify.</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Limited strength and payload capacity compared to commercial systems.</li> <li>The use of 3D-printed parts may result in lower durability over time.</li> </ul> |

## 2.2 Summmary

Chapter 2, "Literature Review," Review on development of industrial robotic arm by Rahul Gautam

This selective operation robotic control method is need to be overcome the problem such as placing or picking object that at distant from the worker. The robotic arm has been developed successfully as the movement of the robot can be controls precisely. It is expensive to change the cable and therefore the designing to reduce the friction on table, is crucial to increase time between maintenance. Survey on Design and Development of competitive low cost Robot Arm With Four Degrees of Freedom by Ashraf Elfasahany In this paper the representation of the design, development and implementation of robot arm is done, which has the ability to perform simple tasks, such as light material handling. The robotic arm is designed and made from acrylic material where servo motors are used to perform links between arms. The servo motors consist of encoder so that no need to use controller. However, the rotation range of the motor is less than  $180^\circ$  span, which greatly decreases the region reached by the arm and the possible positions. The design of the robot arm was for four degrees of freedom.

## Chapter-3

### Proposed Work

#### 3.1 Block Diagram

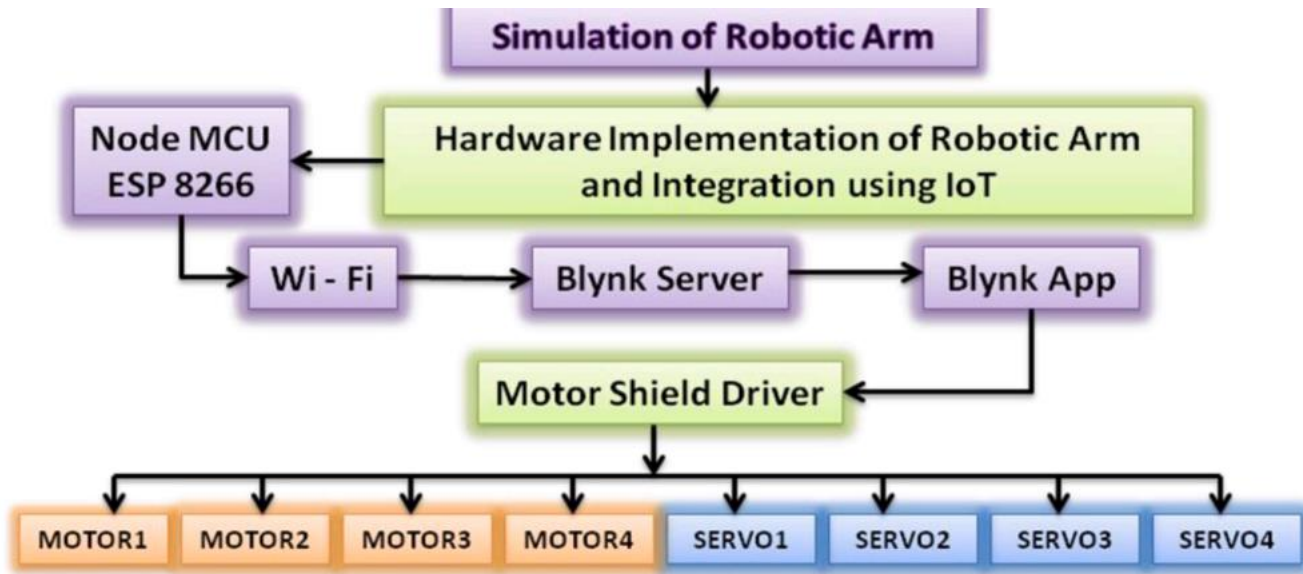


Fig. 1. Block Diagram of the proposed system.



Fig. 2. Integration of Hardware and IoT.

The robotic arm is used for various applications. Yusoff et al. [4] developed a mobile robotic arm, which is wireless and perform functions such as the pick and place function, which is controlled using a PS2 controller. The proposed robotic arm can move to a specific distance, such as moving forward, reverse, turning right and turning left. The mobile robotic is combined with the controller and is based on the Arduino Mega platform and wireless. This arm overcome problems like picking or placing objects very far from the user and picking and placing dangerous objects or things in the easiest and fastest way. A Robotic Arm that can be controlled using the internet was built by Kadir et al. [5]. The robot demonstrates that they can be used within the house for performing human work in day-to-day life. Arduino UNO [6] was used to control the interfaced robot and combined with the internet by employing Arduino Ethernet Shield. The test results showed that the output efficiently controlled between 97% to 99%. Gawli et al. [7], controlled the robot's arm with wire and the internet of things using android mobile remotely.

This robot, control, record movements and perform the same task in repeated mode. Many authors [8–13] represented a controllable robotic arm using the Internet of Things (IoT) technology [7,14–16] considering different degrees of freedom for various applications. In hospitals, nurses can use this Robotic arm to help doctors pick and place objects during surgery, and it can also be useful in the application of taking care of patients [17] by smart phones to control the robotic arm. The gyroscope and Accelerometer will send their signals captured from the android and send it to the Raspberry pi to control the robotic arm. In this work they have emphasised the application of IoT to design a robot arm developed by additive manufacturing with the control interface of Arduino with an HC-05 Bluetooth module. Aisha et al. [18] portrayed that the RoboticArm can be reprogrammed, and its manipulator design is multifunctional to assist and help humans in various works and can be used in factories to pick up and place hands to reduce inefficiency. In automobile industry [19] 3-degree-of- freedom robotic arm along with a mechanical gripper is used. The mean error rate and payload are analysed, and the performance is optimised. In [15], the researchers have discussed hybrid energy system where ESP8266 Wi-Fi module is the main criterion used to module is the main criterion used to switch between the two sources of hybrid energy, wind and solar energy going through any inconvenience through a website. Using IoT, the data transmitted is flexible and is controlled remotely through a web connection with a secured internet. From the Literature we can find that the industrial robot used these days are of programmable type with specific tasks. From the literature we could find that the robots were expensive to manufacture and the programming environment was difficult. Hence, in this work, a robotic arm with 3 DOF independent of each other is simulated and implemented. It is controlled from remote location through IoT from a mobile app. An attempt has been made to integrate as shown in Fig. 2 and the results are compared. The entire robotic arm is designed using basic electronic components and 3D printed manipulating with servo motors.

The robotic arm works on the principle of electrical input energy to perform some mechanical works effectively with the help of some automation and program based operations. The pick and place robotic arm consist of major hardware components such as strips & motors and arm gripper, switches, battery, piece of metal, and other discrete mechanical and electrical components. This project is designed for developing a pick and place robotic arm with a soft catching gripper. This soft catching gripper is used for safely handling an object carefully while Catching and placing.

### 3.2 Summary

Robotic arms, aptly named because they resemble a human arm, are typically mounted to a base. The arm contains multiple joints that act as axes that enable a degree of movement. The higher number of rotary joints.

a robotic arm features, the more freedom of movement it has. A pick and place robotic arm is a type of industrial robot designed to automate the process of picking up objects from one location and placing them in another with high precision and efficiency. These robotic arms are commonly used in manufacturing, assembly, packaging, and material handling industries. They consist of components such as actuators, sensors, end-effectors (grippers), and a control system that coordinates their movement. The robotic arm operates through a combination of pre-programmed instructions and real-time adjustments, often guided by sensors and cameras for accurate object detection and positioning. Its design mimics a human arm, with multiple joints and degrees of freedom, enabling it to perform tasks in confined or repetitive environments where human involvement might be inefficient or unsafe. By integrating technologies such as artificial intelligence, machine vision, and advanced controllers, modern pick and place robots are highly adaptable, capable of handling various object shapes, sizes, and weights with minimal human intervention, thereby improving productivity and consistency in industrial operations.



## Chapter-4

### Hardware Description

#### 4.1 Description of Components

##### 4.1.1 Arduino Uno:

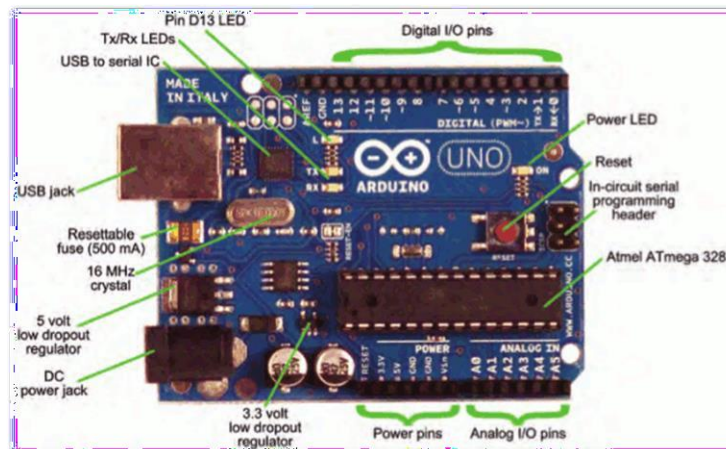


Fig 2: Arduino Uno

Fig 2 the Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller (MCU). The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits.

The Arduino Uno has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery. The microcontroller operates at a voltage of 5V, and the recommended input voltage is 7-12V. It has a flash memory of 32 KB, SRAM of 2 KB, and EEPROM of 1 KB. The clock speed is 16 MHz.

Another crucial role of the Arduino Uno is to facilitate communication and integration among various hardware components within the automated grading system. Through serial communication protocols such as UART or I2C, the Arduino Uno exchanges data with peripheral devices, including the conveyor belt controller, YOLOv5-enabled computing platform, and user interface modules. By acting as a central processing unit, the Arduino Uno orchestrates the synchronized operation of different subsystems, ensuring seamless coordination and synchronization throughout the grading process. Additionally, the Arduino Uno can be interfaced with external devices such as display screens, indicator lights, and feedback mechanisms to provide real-time feedback to operators and enhance system usability.



#### 4.1.2 SG90 Microservo



Servo motors are devices that can rotate to a specific angle or position. They can be used to move robotic arms, steering wheels, camera gimbals, etc. Servo motors have three wires: power, ground and signal. The power wire is usually red and should be connected to the 5V pin on the Arduino board. The ground wire is usually black or brown and should be connected to a ground pin on the board. The signal wire is usually yellow or orange and should be connected to a PWM pin on the board.

The angle is determined by the duration of a pulse that is applied to the control wire. This is called Pulse width Modulation.

The servo expects to see a pulse every 20 ms. The length of the pulse will determine how far the servo turns. For example, a 1.5ms pulse will make the servo turn to the 90 degree position (neutral position).

When a pulse is sent to a servo that is less than 1.5 ms, the servo rotates to a position and holds its output shaft some number of degrees counterclockwise from the neutral point.

The SG90 micro servo motor works by using a microcontroller to send pulse width modulation (PWM) signals to a servo's embedded board:

1. The microcontroller sends PWM signals to the servo's embedded board.
2. The embedded board receives the signals and controls the servo's motor to turn.
3. The motor drives the gear system, which then moves the shaft.
4. The shaft is connected to a potentiometer, which outputs a voltage signal to the embedded board.
5. The embedded board uses the voltage signal to determine the direction and speed of the rotation.

#### 4.1.3 HC05 wireless bluetooth



The HC-05 is a class 2 Bluetooth module designed for transparent wireless serial communication. It is pre-configured as a slave Bluetooth device. Once it is paired to a master Bluetooth device such as PC, smart phones and tablet, its operation becomes transparent to the user. All data received through the serial input is immediately transmitted over the air. When the module receives wireless data, it is sent out through the serial interface exactly as it is received. No user code specific to the Bluetooth module is needed at all in the user microcontroller program. The HC-05 supports two work modes: Command and Data mode. The work mode of the HC-05 can be switched by the onboard push button. The HC-05 is put in Command mode if the push button is activated. In Command mode, user can change the system parameters (e.g. pin code, baud rate, etc) using host controller itself of a PC running terminal software using a serial to TTL converter. Any changes made to system parameters will be retained even after power is removed. Power cycle the HC-05 will set it back to Data Mode. Transparent UART data transfer with a connected remote device occurs only while in Data Mode.

##### **Features:**

- Bluetooth v2.0+EDR
- 2.4GHz ISM band frequency
- Supported baud rate: 9600 (default), 19200, 38400, 57600, 115200, 230400, 460800.
- Speed: Asynchronous: 2.1Mbps(Max) / 160 kbps, Synchronous: 1Mbps/1Mbps
- Power supply: 3.6V to 6V DC

#### 4.1.4 MG996R Gear servo



MG996R servo motor is a high-torque and precise motor ideal for robotics and automation projects. With its durable construction, strong torque of up to 11 kg/cm, and quick response time, it provides reliable and accurate motion control. Compatible with various control systems, this servo motor offers flexibility and ease of integration into your projects. This High-Torque MG996R Digital Servo features metal gearing resulting in extra high 10kg stalling torque in a tiny package. The MG996R is essentially an upgraded version of the famous MG995 servo, and features upgraded shock-proofing and a redesigned PCB and IC control system that make it much more accurate than its predecessor. The gearing and motor have also been upgraded to improve dead bandwidth and centering. The unit comes complete with 30cm wire and 3 pin 'S' type female header connector that fits most receivers, including Futaba, JR, GWS, Cirrus, Blue Bird, Blue Arrow, Corona, Berg, Spektrum and Hitec.

##### Basic Specifications:

- **Type:** Digital servo motor (gear motor).
- **Voltage:** Operates typically at **4.8V to 7.2V**.
- **Torque:**
  - At **4.8V**, the MG996R provides a maximum torque of **9.4 kg·cm (130 oz·in)**.
  - At **6.0V**, the maximum torque is **11 kg·cm (152 oz·in)**.
- **Speed:**
  - At **4.8V**, it has a speed of **0.2 seconds per 60°** rotation.
  - At **6.0V**, the speed is around **0.15 seconds per 60°**.
- **Rotation Range:** Standard servos like the MG996R typically rotate **180°** (i.e., 90° in each direction from the center).
- **Weight:** The servo weighs around **55 grams (1.94 oz)**.
- **Dimensions:** About **40.7 x 19.7 x 42.9 mm** (L x W x H).

#### 4.1.5 Resistors



A resistor is an electrical component that limits or regulates the flow of electrical [current](#) in an electronic circuit. Resistors can also be used to provide a specific voltage for an active device such as a [transistor](#). The most common type in electronic devices and systems is the *carbon-composition resistor*. Fine granulated carbon (graphite) is mixed with clay and hardened. The resistance depends on the proportion of carbon to clay; the higher this ratio, the lower the resistance.

Resistors have a wide variety of uses in electronic circuits, including:

- **Current Limiting:** Protects components like LEDs by limiting the current passing through them.
- **Voltage Division:** Used in voltage divider circuits to split the voltage between components.
- **Signal Filtering:** In combination with capacitors and inductors, resistors help filter signals in audio systems, radios, etc.
- **Heat Generation:** In power resistors, they convert electrical energy into heat, which can be useful in certain heating applications.

Resistors are essential components in electronic circuits, providing controlled resistance to electrical current. Their variety in types, resistance values, and applications makes them indispensable in almost all electrical and electronic devices, from simple household gadgets to complex systems like computers and telecommunications.

#### 4.1.6 Laptop:



A **laptop** is a portable computer that integrates all the essential components of a desktop computer into a compact, lightweight, and battery-powered form. Laptops are commonly used for personal, professional, educational, and entertainment purposes due to their mobility, versatility, and ability to run software applications similar to desktop computers.

##### **Central Processing Unit (CPU):**

- The CPU is the brain of the laptop, responsible for executing instructions and performing calculations.  
This laptop may serve several purposes:
- **Control Unit:** It could act as the control center for the robotic arm, sending commands and receiving feedback.
- **Data Processing Unit:** The laptop may process sensor data, video feeds, and other inputs from the robotic arm.
- **Monitoring Unit:** The laptop could display real-time status information, alerts, and visual feedback from cameras attached to the robotic arm.

#### 4.1.7 Jumper Wires:



Fig is the jumper wires seamlessly integrate with the Arduino Uno microcontroller and peripheral devices, serving as the primary means of communication and signal routing between various hardware modules. Whether connecting sensors for data acquisition, interfacing actuators for control, or linking display modules for user feedback, jumper wires play a crucial role in establishing reliable electrical connections that enable the seamless operation of the grading system. Through careful organization and labeling of jumper wire connections, operators can ensure clarity and ease of maintenance, simplifying troubleshooting and system maintenance tasks throughout the lifecycle of the automated grading system.

#### **4.1.8 Power Supply Adapter:**



Power Supply Adapter

The Power Supply Adapter, rated at 12V and 2A, is a critical component in the automated grading system for agricultural products, providing the necessary electrical power to drive and operate various hardware components. With its output voltage of 12 volts and current capacity of 2 amps, this power adapter delivers a total power output of 24 watts, ensuring sufficient power supply for the operation of the conveyor belt, motor drivers, microcontrollers, sensors, and other peripheral devices within the grading system. Equipped with a standard DC barrel connector or terminal block interface, the power supply adapter offers compatibility and ease of integration with the electrical infrastructure of the grading setup.

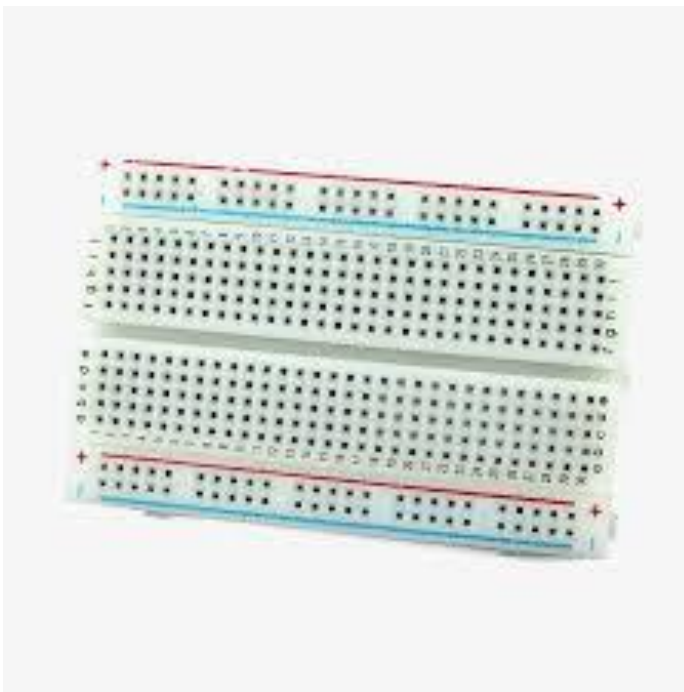
#### 4.1.9 Arduino USB cable:



Arduino USB cable

Fig 12 the Arduino USB cable facilitates bidirectional data transfer between the Arduino Uno microcontroller and the connected computer or programming device. Through this cable, operators can upload firmware, sketches, and code libraries to the Arduino Uno, enabling customization and configuration of the grading system's behavior and functionality. Additionally, the USB connection allows for real-time monitoring and debugging of the grading process, providing operators with insights into system performance, sensor readings, and debugging information through the Arduino Integrated Development Environment (IDE) or serial communication tools.

#### 4.1.10 breadboard



A breadboard (also known as a protoboard) is a tool used for constructing electronic circuits without soldering. It is a reusable platform that allows components to be quickly and easily connected together to form circuits. Breadboards are especially useful for prototyping, testing, and debugging electronic circuits before finalizing designs for permanent soldered boards, such as printed circuit boards (PCBs).

### Structure of a Breadboard

A typical breadboard consists of a grid of holes into which electronic components (such as resistors, capacitors, diodes, transistors, and ICs) can be inserted. These holes are electrically connected in rows and columns, which makes it possible to establish connections between components without needing to solder.

### How Breadboards Work

**Electrical Connections:** Each row (except for power rails) is internally connected. For example, a set of 5 holes in a row (typically) will be connected together, and components inserted into these holes will be electrically connected.

**Component Placement:** Components such as resistors, capacitors, ICs, and wires are placed into these holes to form an electrical path. The breadboard allows for easy rearrangement of components, which is crucial for debugging or modifying a circuit.

**Jump Wires:** Connections between rows and columns can be made using jumper wires. These wires are often used to bridge gaps or extend connections across different parts of the breadboard. A jumper wire might connect the positive rail to a component or connect two components that are far apart on the board.



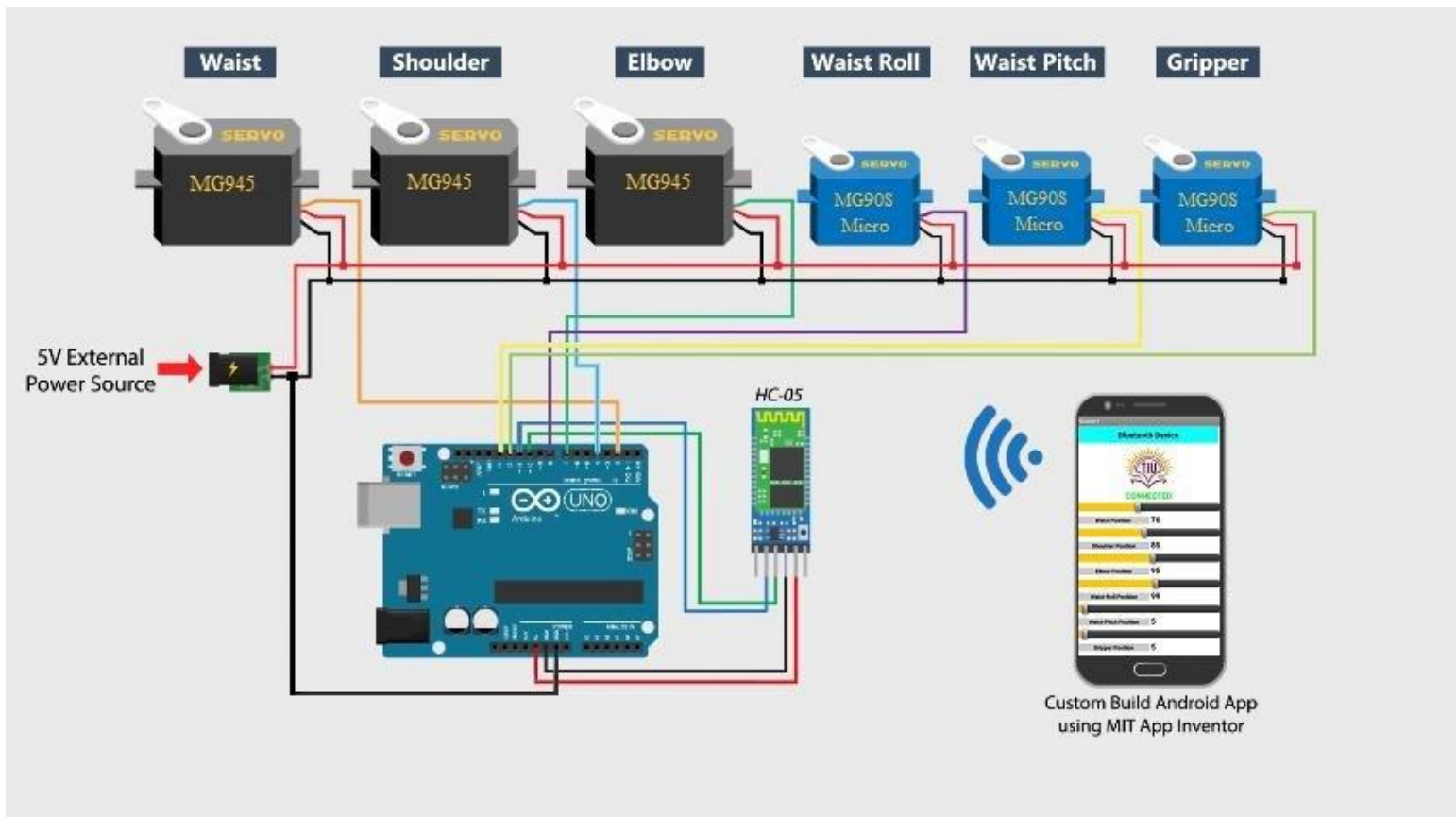
## 4.2 Summary

Chapter 4 of A **pick and place robotic arm** involves multiple integrated components that work together to provide precise control, remote operation, and continuous monitoring. The core components include the robotic arm's actuators, sensors, and controllers, along with a laptop or central control unit that communicates via wireless networks. IoT integration adds the ability for remote control, cloud-based data storage, real-time feedback, and enhanced system functionality through AI/ML algorithms. This combination of hardware and software allows robotic arms to perform complex tasks autonomously or with minimal human intervention, making them ideal for industries like manufacturing, healthcare, logistics, and research. Although research on robotic arm has been conducted in the past, the lack of technology applied has given the chance to conduct this research by implementing IoT for the robotic arm. Android mobile device is utilized as the platform to remotely controlled this robotic arm to perform the pick and place operations. Arduino Yun is used as the brain of the system, which will process the control signal and output modulation signal to the servo motor.

## Chapter 5

### Implementation

#### 5.1 Hardware implementation



**Circuit diagram of pick and place robotic arm using iot**

The hardware implementation of a pick-and-place robotic arm involves assembling and integrating various mechanical, electrical, and electronic components to create a functional system. The robotic arm will be designed to pick up objects and place them at a desired location, using motors, sensors, a micro controller, and other components. The Robotic Arm implemented in our study is a 3D printer Robotic Arm with 3 Degrees of Freedom (DOF).[5,20–22] It has 3 servo motors attached at 3 different joints of the Robotic Arm and 1 Gripper as the end effectors. The initial design is implemented.

Integrating electrical components like motors and sensors, wiring them correctly to the microcontrolled programming the system to perform pick-and-place tasks. By using motors, sensors, and a microcontroller, the arm can be controlled to pick up objects and place them in specific locations based on the programmed logic. The process requires both mechanical assembly skills and electrical wiring, along with software programming to control the arm's movements and behaviors.

## 5.2 Software Implementation

### Flowchart

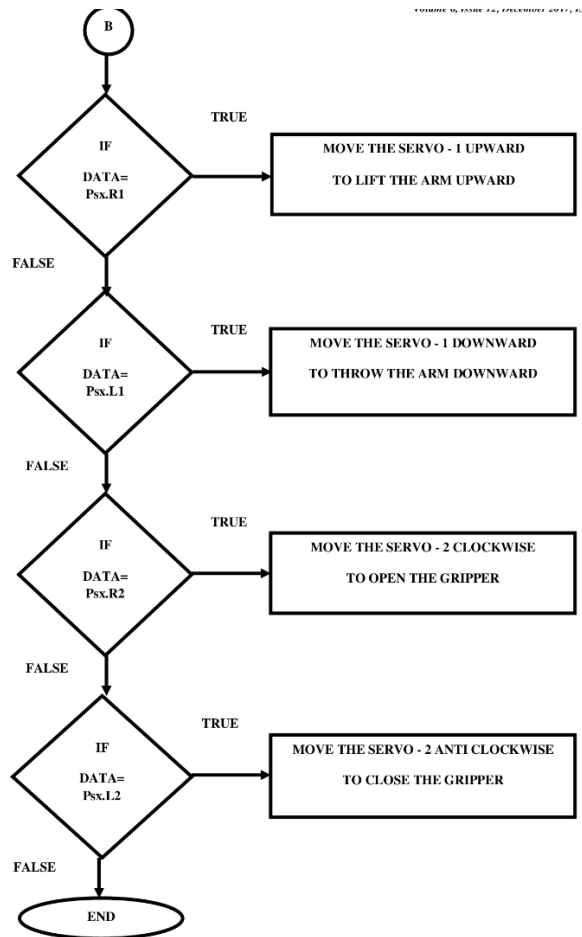
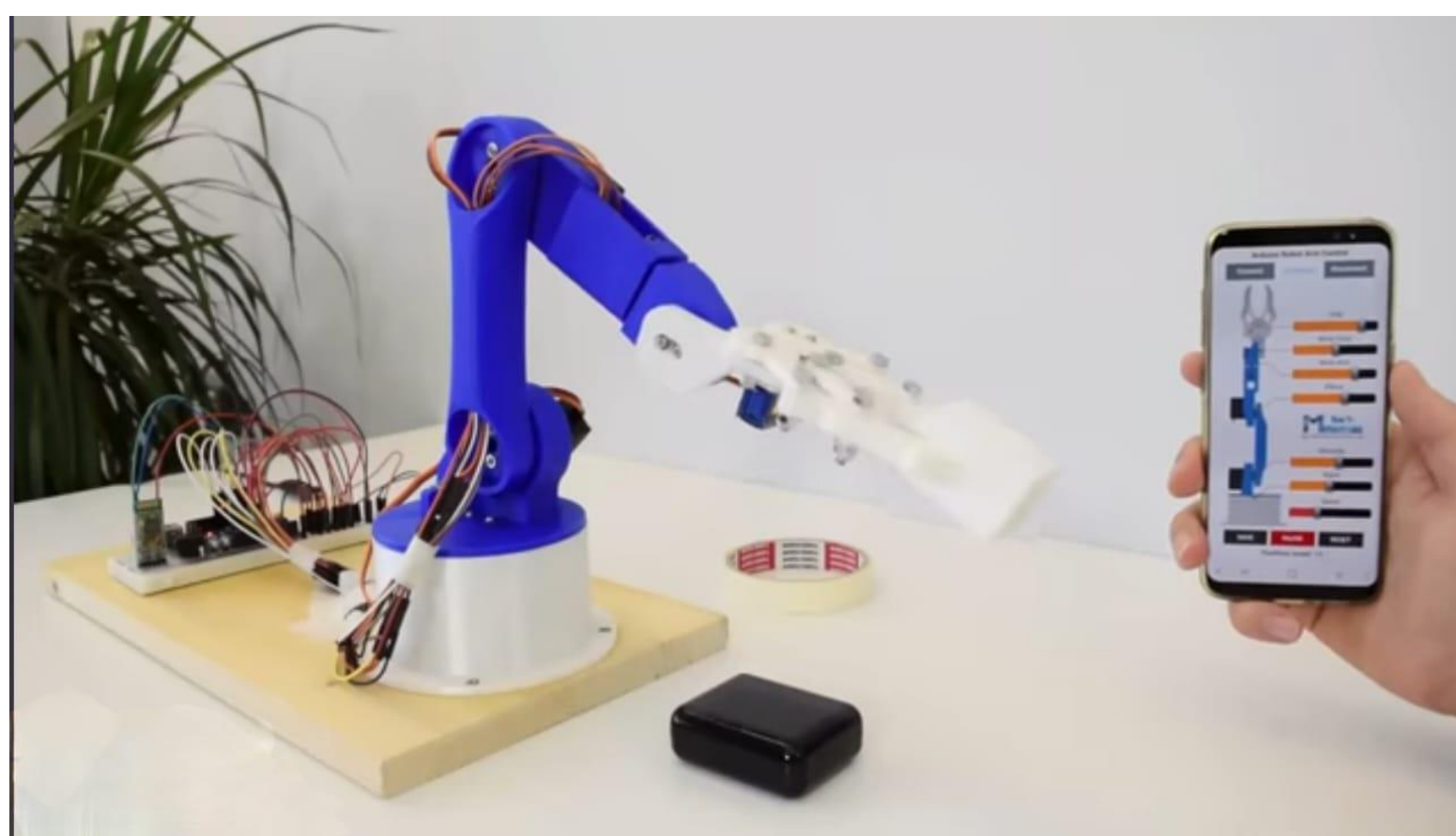


Fig : Flowchart

Software programming involves the use of computer language to develop a sequence or a set of Instructions that will carry out a specific task. It plays a significant role by providing the necessary platform in controlling the action and the movement of the robotic arm. the flow chart for robotic arm control using IoT. During automatic mode, the robotic arm picks an item from the first position and release the item in the second position. The process of pick and place operation is then repeated. During manual mode, the robotic arm is train through android mobile device.

If your robot arm needs to pick objects autonomously, a vision system can be used to detect and locate objects. For a simple implementation, you might use a **camera** and a **color detection** algorithm to find objects. This implementation outlines a simple pick and place system. More sophisticated systems would involve precise control algorithms, real-time sensor data integration, and more advanced motion planning. accordingly.



The working principle of a pick-and-place robotic arm involves automating the process of picking up objects from one location and placing them at another, typically using a combination of sensors, actuators, and control systems. The main steps are:

1. **Object Detection:** Sensors (like cameras or proximity sensors) or vision systems identify the position and orientation of the object to be picked.
2. **Gripping the Object:** The robotic arm's gripper (or end effector) moves to the object, adjusts for its position, and securely picks it up using mechanical jaws, vacuum suction, or magnets, depending on the type of gripper.
3. **Movement to Destination:** The robot moves the object along a predefined or dynamically adjusted path, often using motors and actuators controlled by a central controller. The arm follows specific motion instructions, such as Cartesian or articulated movements, to avoid obstacles.
4. **Placing the Object:** The robot positions the object at the desired location and releases it by opening the gripper or deactivating the vacuum/magnetic grip.
5. **Feedback and Adjustment:** Sensors provide feedback to adjust grip strength or movement accuracy to ensure correct object handling and placement.

The robotic arm's movement and actions are governed by algorithms like inverse kinematics (IK) to ensure precise positioning and task execution.

## Torque

To calculate the *torque required* for each servo motor in your robotic arm, you need to consider the weight of the arm segments and any objects it might lift, as well as the length of each arm segment. Torque is calculated as:

$$\text{Torque (T)} = \text{Force (F)} \times \text{Distance (d)}$$

Where:

- *Force (F)* is the weight (mass  $\times$  gravity,  $(F = m \times g)$ , where  $(g = 9.8 \text{ m/s}^2)$ ).
- *Distance (d)* is the perpendicular distance from the axis of rotation to the line of action of the force.

### Steps to Determine Torque for Each Joint:

#### 1. *Identify Load (Force) on Each Joint:*

- Add the weight of the segment *beyond the joint* and any additional load (e.g., the gripper and object weight).

#### 2. *Calculate Distance (Lever Arm):*

- Measure the length of each arm segment from the axis of rotation (joint center) to the end of the segment.

#### 3. *Account for Multiple Segments:*

- The torque required at each joint includes the sum of the torques caused by all segments and weights beyond that joint.

The torque required for a pick and place robotic arm will vary based on the size, weight, and design of the system. By accurately calculating the torque needed at each joint and selecting appropriate motors, you can ensure the arm operates efficiently and performs tasks with the required precision and speed.

## PROGRAM:

```
# Servo Motor Specifications
servo_motors = {
    "MG996R": {"torque_per_motor": 9.4, "count": 3}, # kg·cm
    "SG90": {"torque_per_motor": 1.8, "count": 3}, # kg·cm
}
```

```
# Load distribution (distance and weight in cm and kg)
load_distribution = {
    "Waist": {"distance": 5.0, "weight": 0.8},    # distance in cm, weight in kg
    "Shoulder": {"distance": 4.0, "weight": 0.6},
    "Elbow": {"distance": 3.0, "weight": 0.5},
    "Wrist Roll": {"distance": 2.0, "weight": 0.3},
    "Wrist Pitch": {"distance": 1.5, "weight": 0.2},
    "Gripper": {"distance": 1.0, "weight": 0.1},
}

# Gravitational acceleration
g = 9.8 # m/s^2

# Calculate the torque required for each joint
def calculate_load_torque(load_distribution):
    total_torque = 0
    individual_torques = {}
    for joint, specs in load_distribution.items():
        # Torque = weight (kg) * g (m/s^2) * distance (cm) / 100 (to convert cm to m)
        torque = specs["weight"] * g * (specs["distance"] / 100) # torque in kg·cm
        individual_torques[joint] = torque
        total_torque += torque
    return total_torque, individual_torques

# Calculate torque utilization per servo type
def calculate_total_torque_with_weight(servo_motors, total_torque):
    utilization = {}
    for servo, specs in servo_motors.items():
        total_motor_torque = specs["torque_per_motor"] * specs["count"]
        utilization[servo] = {
            "total_motor_torque": total_motor_torque,
            "remaining_torque": max(total_motor_torque - total_torque, 0)
        }
    return utilization

# Perform the calculations
total_torque, individual_torques = calculate_load_torque(load_distribution)
torque_utilization = calculate_total_torque_with_weight(servo_motors, total_torque)

# Display Results
print("Torque Breakdown by Joint (Including Object Weight):")
for joint, torque in individual_torques.items():
    print(f"{joint}: {torque:.2f} kg·cm")

print(f"\nTotal Torque Required (Including Object Weight): {total_torque:.2f} kg·cm\n")

print("Torque Utilization (Including Object Weight):")
for servo, details in torque_utilization.items():
    print(f"{servo} - Total Torque: {details['total_motor_torque']} kg·cm, Remaining Torque: {details['remaining_torque']} kg·cm")
```

## Chapter 6

### CONCLUSION

This report presents the design and the development of robotic arm, which has the ability to perform simple tasks, such as light material handling. The robotic arm is designed and built from aircraft grade aluminium material where servo motors were used to perform arm movements. The design of the robotic arm limited to the four degrees of freedom The design of a Robotic arm has been complete. A prototype was built and confirmed functional. This system would make it easier for man to unrivalled the risk of handling suspicious objects which could be hazardous in its present environment and workplace. Complex and complicated duties would be achieved faster and more accurately with this design.

Although research on robotic arm has been conducted in the past, the lack of technology applied has given the chance to conduct this research by implementing IoT for the robotic arm. Android mobile device is utilized as the platform to remotely controlled this robotic arm to perform the pick and place operations. Arduino Yun is used as the brain of the system, which will process the control signal and output modulation signal to the servo motor. Future research needs to examine the effects of other types of sensor and actuator. Radio control servo motor with greater angle of rotation can be used to improve the reachability of the robotic arm. Moreover, more advance motor such as stepper motor can be used to offer more accurate and precise movement. This future research could help in the review of available robotic arm in the industries and make recommendations for future editions. Finally, future research and development in this research is greatly.

The pick and place robotic arm project highlights the potential of automation in simplifying repetitive tasks, improving efficiency, and reducing human intervention. By utilizing an IoT-based system controlled through an Android mobile device and powered by an Arduino Yun, the robotic arm successfully performs pick and place operations with precision and reliability. This system demonstrates the practical application of modern technologies in robotics, allowing for remote control and monitoring of the robotic arm from anywhere. While the current system proves effective, there are opportunities for further enhancements. The integration of more advanced sensors, such as vision or proximity sensors, could enable more accurate object detection and handling. Additionally, upgrading the actuators to more precise and powerful motors, like stepper motors, could improve the arm's performance, providing greater accuracy and flexibility in operations.

## 6.1 Recommendations for future research and Improvements

### 1. Incorporation of Advanced Sensors

#### a. Vision Systems:

- **Current Limitation:** Many robotic arms rely on basic sensors or predefined positions for picking objects. These methods are limited in dynamic environments where objects may be misplaced or have varying shapes and sizes.
- **Future Scope:** Integrating vision systems, such as cameras combined with machine learning algorithms (e.g., object recognition and tracking), will allow the robotic arm to "see" and identify objects in real-time. This would enable the robotic arm to handle more complex tasks in unstructured environments, such as sorting objects by color, shape, or size.

#### b. Force/Torque Sensors:

- **Current Limitation:** Basic pick and place arms can sometimes damage delicate objects or fail to handle them precisely due to lack of real-time feedback.
- **Future Scope:** Force sensors could be integrated to provide feedback during the interaction with objects, allowing for delicate handling or adaptive force control, preventing damage to fragile items. These sensors could enable the robotic arm to sense and adjust its grip or placement force automatically.

#### c. Proximity and Distance Sensors:

**Future Scope:** Adding proximity or ultrasonic sensors would allow the robotic arm to sense objects in its path, enabling safer, more efficient operation in shared spaces with human workers.

### 2. Enhanced Actuators and Motors

#### a. Stepper Motors and Servo Motors:

- **Current Limitation:** Traditional servo motors may not provide sufficient precision or torque in certain applications.
- **Future Scope:** Upgrading to stepper motors or DC motors with higher precision and torque will allow for finer control and smoother movements. Stepper motors offer high accuracy without the need for additional encoders, making them ideal for applications requiring precise movement.

#### b. Pneumatic and Hydraulic Actuators:

- **Future Scope:** Incorporating pneumatic or hydraulic actuators could enhance the strength and flexibility of robotic arms, enabling them to handle heavier or more delicate items with improved speed and force. These actuators are particularly useful in high-load, high-speed applications like assembly lines.

#### c. Soft Robotics:

- **Future Scope:** Soft robotic actuators that mimic human-like flexibility could be used in applications requiring gentle handling, such as in healthcare (handling surgical instruments or delicate items in pharmaceuticals).



### 3. Artificial Intelligence (AI) and Machine Learning

#### a. Autonomous Operation:

- **Current Limitation:** Most robotic arms still require manual control or pre-programming for specific tasks.
- **Future Scope:** AI can be integrated into the robotic arm to allow it to make real-time decisions, such as dynamically adjusting its actions based on sensory feedback, changing environments, or performing predictive maintenance. AI can help the robotic arm learn from its mistakes, adapt to new situations, and optimize its operations.

#### b. Predictive Analytics:

- **Future Scope:** By integrating machine learning algorithms with sensors, the robotic arm can predict the required movements based on historical data, environmental conditions, and load patterns, leading to higher operational efficiency. This would be especially useful in high-volume operations such as manufacturing.
- 

### 4. Collaborative Robots (Cobots)

#### a. Human-Robot Collaboration:

- **Current Limitation:** Traditional robotic arms are typically used in isolated environments due to safety concerns when working alongside humans.
  - **Future Scope:** Cobots, or collaborative robots, are designed to work safely alongside human workers. Future research should focus on improving the safety protocols and sensors (e.g., pressure-sensitive joints, collision avoidance, and real-time adaptive behavior) that allow the robotic arm to work in proximity to humans without causing harm. This can enable more flexible and adaptive workflows, especially in industries like healthcare and small-scale manufacturing.
- 

### 5. Mobility and Flexibility

#### a. Mobile Robotic Arms:

- **Current Limitation:** Many robotic arms are stationary, limiting their functionality in environments where the location of the arm or objects changes frequently.
- **Future Scope:** Implementing mobile robotic platforms (e.g., autonomous wheeled or tracked robots) that carry the robotic arm would increase its versatility. This allows the robotic arm to navigate through dynamic environments, pick and place objects from various locations, or perform a variety of tasks in warehouses or factories.

#### b. Multi-Arm Systems:

- **Future Scope:** Integrating multiple robotic arms that can collaborate in performing tasks simultaneously could enhance efficiency in complex pick and place operations. These multi-arm systems could work in tandem to handle larger or more intricate items or increase the speed of operations in industrial settings.
-

## 6. Integration with IoT and Cloud Computing

### a. Real-Time Monitoring and Diagnostics:

- **Future Scope:** By integrating IoT capabilities, the robotic arm can be connected to a network for real-time monitoring, diagnostics, and performance analysis. This would enable predictive maintenance, where the system can notify operators when parts are likely to wear out or fail, reducing downtime and improving overall efficiency.

### b. Cloud-Based Control Systems:

- **Future Scope:** Cloud integration can allow remote control and monitoring of the robotic arm from anywhere. This would enable manufacturers to deploy robotic arms in distributed environments, optimize their operations remotely, and even share data between multiple robots to improve their efficiency and coordination.
- 

## 7. Customization and Scalability

### a. Modular Design:

- **Future Scope:** Designing robotic arms with modular components will make them more customizable and adaptable to different applications. The ability to easily swap out parts such as grippers, sensors, or motors will allow businesses to tailor the robotic arm to their specific needs, reducing costs and increasing flexibility.

### b. Scalability for Industry-Specific Applications:

- **Future Scope:** The development of customizable pick and place robotic arms can be extended to a wide range of industries, such as pharmaceuticals (for handling fragile medicines), food (for picking delicate fruits or packaging), or even healthcare (handling medical supplies). Each industry can benefit from the scalability of robotic arms designed with specific needs in mind.
- 

## 8. Sustainability and Energy Efficiency

### a. Green Technologies:

- **Future Scope:** Future pick and place robotic arms should be designed with energy efficiency in mind. This includes using low-power motors, implementing regenerative braking systems, and ensuring that the robotic arm uses minimal energy during idle times. Sustainable materials and eco-friendly designs could also be considered to reduce environmental impact.
- 

## Conclusion:

The future of pick and place robotic arms is bright, with numerous avenues for technological enhancements and advancements. By incorporating AI, advanced sensors, more efficient actuators, and collaborative features, robotic arms will become more intelligent, versatile, and capable of performing tasks in increasingly complex environments. The integration of IoT and cloud computing will provide real-time monitoring, predictive maintenance, and remote control, making these systems more efficient and accessible. As industries continue to seek ways to improve automation, the evolution of pick and place robotic arms will play a pivotal role in shaping the future of manufacturing, logistics, healthcare, and more.

## 6.2 Future scope in various fields

### 1. Manufacturing and Assembly

#### a. Mass Production:

- **Future Scope:** In high-volume production environments, robotic arms will be capable of handling a diverse range of materials and products with more speed, precision, and flexibility. Advanced vision systems combined with AI will enable robots to inspect parts, adjust assembly line processes, and optimize production workflows dynamically.
- **Integration:** These robotic systems will likely be integrated with automated quality control systems, reducing human error and maintaining consistent product standards.

#### b. Customization in Production:

- **Future Scope:** With modular and flexible designs, pick and place robots will be able to quickly switch between tasks, allowing for more customized, small-batch production runs. This is ideal for industries that require the manufacture of customized or complex products on-demand, such as automotive or electronics.

### 2. Logistics and Warehousing

#### a. Smart Warehouses:

- **Future Scope:** With the advent of AI, machine learning, and real-time data analytics, pick and place robots will be pivotal in creating fully automated warehouses. These systems will be able to navigate complex environments, select, move, and organize goods without human intervention. The use of collaborative robots (cobots) will further enhance efficiency by allowing robots to work alongside human workers in handling diverse tasks.

#### b. Autonomous Mobile Robots (AMRs):

- **Future Scope:** Integration of pick and place robotic arms with autonomous mobile robots will revolutionize warehouses by enabling robots to move throughout the facility, pick up goods from shelves, and transport them to packing or sorting areas. This will improve the speed and efficiency of order fulfillment, reduce labor costs, and streamline logistics operations.

---

### 3. Healthcare and Pharmaceuticals

#### a. Handling Medical Supplies:

- **Future Scope:** In healthcare, pick and place robotic arms could be used to handle and organize medical instruments and supplies, reducing contamination risks. These robots can perform tasks such as picking up clean surgical tools, placing them in sterilized environments, or organizing medications.
- **Precision and Sterility:** Robotic arms with enhanced precision, force sensors, and vision systems could be employed for complex tasks, such as sorting medications, handling sensitive materials, or performing simple medical procedures, like filling syringes or preparing doses.

#### b. Pharmacy Automation:

- **Future Scope:** In pharmacies, pick and place robotic arms will play a critical role in automating the process of selecting, sorting, and dispensing medications. Robots could be used in high-accuracy environments to handle medication bottles, manage inventory, or package and label products, making pharmaceutical operations more efficient and error-free.
-

- **4. Food and Beverage Industry**

- **a. Sorting and Packing:**

- **Future Scope:** In food processing plants, pick and place robotic arms can be used for sorting fruits, vegetables, or other food items based on size, weight, or ripeness. Using AI and advanced vision systems, the robots can even identify damaged or spoiled items and remove them from production lines, ensuring quality control.
- **Precision and Hygiene:** Pick and place robots will evolve to work in hygiene-critical environments, handling food products in a way that minimizes contamination risks while maintaining precision in packaging and placement.

- **b. Automated Cooking and Meal Preparation:**

- **Future Scope:** Robotic arms could move from handling ingredients to fully preparing meals, picking up ingredients, assembling dishes, and packaging them for delivery. This would transform industries such as catering, meal preparation services, and even the fast-food sector.
- 

- **5. Retail**

- **a. Stocking and Restocking Shelves:**

- **Future Scope:** In retail, robotic arms will be used to automate the stocking and restocking of shelves. They will have the ability to recognize items in low-stock areas, retrieve inventory from storage, and place them on shelves. AI and machine learning will ensure that robots can recognize product packaging and barcode labels accurately, even in cluttered environments.

- **Customer Interaction:** In the future, robots could assist customers directly in stores by providing product information, helping them pick products, or even transporting items to checkout areas.

- **b. Automated Checkout:**

- **Future Scope:** With robotic arms integrated with cashier-less checkout systems, robots could handle the packing and scanning of products in real-time. This will streamline the retail experience, reduce wait times, and enhance operational efficiency.
- 

- **6. Agriculture**

- **a. Harvesting and Sorting:**

- **Future Scope:** Pick and place robotic arms in agriculture can automate the harvesting of fruits, vegetables, and other crops. With the addition of vision and force sensors, robotic systems could pick delicate fruits or vegetables without damaging them, sort them by size, and prepare them for packaging.

- **Precision Agriculture:** By integrating IoT and AI, these robotic arms can monitor crop health, detect pests, or measure growth, ensuring that crops are harvested at optimal times and contributing to sustainable farming practices.
-

- **b. Automated Greenhouse Management:**

- **Future Scope:** In greenhouses, robotic arms could be used for a variety of tasks such as planting, pruning, and watering plants, enhancing agricultural efficiency and minimizing the need for human labor in harsh or dangerous conditions.
- 

- **7. Education and Research**

- **a. Prototyping and Experimentation:**

- **Future Scope:** In research labs, pick and place robotic arms could assist with repetitive tasks, such as transferring test tubes, organizing experiments, or assembling complex prototypes. Their precision would ensure that researchers can focus on more complex tasks while the robot handles routine functions.
  - **STEM Education:** Robotics systems will become an essential tool for teaching in educational settings. Pick and place robots can be used in classrooms to teach students about automation, programming, and robotics through hands-on activities, fostering interest in STEM fields.
- 

- **8. Space and Exploration**

- **a. Robotic Assistance in Space Missions:**

- **Future Scope:** Pick and place robotic arms are already used on space stations and spacecraft to handle tools, samples, and other critical materials. In the future, these robotic arms could assist astronauts with complex tasks, such as assembling parts, conducting experiments, or managing equipment in space environments.
  - **Extraterrestrial Exploration:** On Mars or the Moon, robotic arms will play a crucial role in handling soil samples, conducting construction, and assembling infrastructure for human settlements.
- 

- **9. Environmental and Recycling**

- **a. Waste Sorting:**

- **Future Scope:** Pick and place robotic arms can be used to improve recycling operations by automatically sorting waste materials such as plastics, metals, and papers based on size, shape, or type. The integration of AI and advanced sensors will enable the robotic system to identify and sort recyclable materials with greater efficiency.
  - **Environmental Cleanup:** Robotic arms could also assist in environmental cleanup operations, removing harmful debris or pollutants from oceans, rivers, or other contaminated areas.
-

## 6.4 Summary

Chapter 7 of A **pick and place robotic arm** involves multiple integrated components that work together to provide precise control, remote operation, and continuous monitoring. The core components include the robotic arm's actuators, sensors, and controllers, along with a laptop or central control unit that communicates via wireless networks. IoT integration adds the ability for remote control, cloud-based data storage, real-time feedback, and enhanced system functionality through AI/ML algorithms. This combination of hardware and software allows robotic arms to perform complex tasks autonomously or with minimal human intervention, making them ideal for industries like manufacturing, healthcare, logistics, and research. Although research on robotic arm has been conducted in the past, the lack of technology applied has given the chance to conduct this research by implementing IoT for the robotic arm. Android mobile device is utilized as the platform to remotely controlled this robotic arm to perform the pick and place operations. Arduino Yun is used as the brain of the system, which will process the control signal and output modulation signal to the servo motor.

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## APPENDIX

### Code

```
import serial
import time

# Initialize the serial connection to the Arduino
# Ensure the correct COM port and baud rate are used (e.g., COM3 and 9600)
ser = serial.Serial('COM3', 9600, timeout=1)
time.sleep(2) # Wait for the connection to establish

def send_command(command):
    """
    Send a command to Arduino through serial communication
    :param command: String command to be sent to Arduino
    """
    ser.write(command.encode()) # Send command as bytes
    time.sleep(1) # Delay to ensure the command is processed by Arduino

def pick_object():
    """
    Function to control the robotic arm to pick up an object.
    This will activate the servos for the 'picking' position.
    """
    send_command('PICK') # Sending 'PICK' command to Arduino
    print("Picking up the object...")

def place_object():
    """
    Function to control the robotic arm to place an object.
    This will activate the servos for the 'placing' position.
    """
    send_command('PLACE') # Sending 'PLACE' command to Arduino
    print("Placing the object...")

def reset_arm():
    """
    Reset the arm to the initial position.
    """
    send_command('RESET')
    print("Resetting the arm...")

if __name__ == "__main__":
```



```
try:
    # Command sequence for the pick and place operation
    print("Starting pick and place operation")

    pick_object() # Pick the object
    time.sleep(2) # Wait for the operation to complete

    place_object() # Place the object
    time.sleep(2) # Wait for the operation to complete

    reset_arm() # Reset the arm after task
    print("Pick and place operation completed")

except KeyboardInterrupt:
    print("Program interrupted")

finally:
    ser.close() # Close the serial connection when done

// Define servos for each part of the robot arm
Servo waistServo;
Servo shoulderServo;
Servo elbowServo;
Servo wristRollServo;
Servo wristPitchServo;
Servo gripperServo;

// Bluetooth serial communication
char command;

void setup() {
    // Attach servos to pins
    waistServo.attach(3);
    shoulderServo.attach(5);
    elbowServo.attach(6);
    wristRollServo.attach(9);
    wristPitchServo.attach(10);
    gripperServo.attach(11);

    // Initialize serial communication
    Serial.begin(9600); // For HC-05
}

void loop() {
    if (Serial.available()) {
        command = Serial.read(); // Read the command from the smartphone app
```

```
// Parse the command and move servos accordingly
switch (command) {
  case 'A': // Example command to move waist servo to position 0
    waistServo.write(0);
    break;
  case 'B': // Move waist servo to 90 degrees
    waistServo.write(90);
    break;
  case 'C': // Move shoulder servo to 45 degrees
    shoulderServo.write(45);
    break;
  case 'D': // Move elbow servo to 120 degrees
    elbowServo.write(120);
    break;
  case 'E': // Open gripper
    gripperServo.write(0);
    break;
  case 'F': // Close gripper
    gripperServo.write(90);
    break;
  // Add more cases for additional controls
}
}
```

## Arduino Code

```
#include <Servo.h>

// Define servos for each part of the robot arm
Servo waistServo;
Servo shoulderServo;
Servo elbowServo;
Servo wristRollServo;
Servo wristPitchServo;
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  if (Serial.available()) {
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        break;
      case 'C': // Move shoulder servo to 45 degrees
        shoulderServo.write(45);
        break;
      case 'D': // Move elbow servo to 120 degrees
        elbowServo.write(120);
        break;
      case 'E': // Open gripper
        gripperServo.write(0);
        break;
```

```
case 'F': // Close gripper
    gripperServo.write(90);
    break;
// Add more cases for additional controls
}
}
}
```

|  |                                                                                                                                                | <b>AMC ENGINEERING COLLEGE</b><br><b>DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING</b> |                               |                               |                               |                               |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| <b>PROJECT WORK SELF EVALUATION FORM</b>                                          |                                                                                                                                                |                                                                                                  |                               |                               |                               |                               |
| <b>Name</b>                                                                       |                                                                                                                                                |                                                                                                  |                               | <b>USN</b>                    |                               |                               |
| <b>Program</b>                                                                    |                                                                                                                                                | Bachelor of Engineering in Electronics and Communication                                         |                               |                               |                               |                               |
| S.No                                                                              | To what extent did each of the following you contribute to your project                                                                        | To a very great extent                                                                           | To a great extent             | To a moderate extent          | To some extent                | Not at all                    |
| 1.                                                                                | Have you applied the knowledge of mathematics, science and engineering concepts(PO1,Pso1)                                                      | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 2.                                                                                | Have you analyzed the system of hardware, software and its interaction (PO2,PSO1)                                                              | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 3.                                                                                | Design the systems, components or processes to meet needs(PO3,PSO1)                                                                            | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 4.                                                                                | Have you used the modern tools of engineering to design the system or components of your project (PO5,PSO1)                                    | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 5.                                                                                | Identification of impact in societal and environmental contexts of your project developed in interdisciplinary domain (PO7,PSO2)               | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 6.                                                                                | Are you responsible, diligent, and ethical on your project (core and interdisciplinary) development period with your team members? (PO8, PSO2) | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |
| 7.                                                                                | Have you functioned effectively as a member in a team with your project members? (PO9)                                                         | <input type="checkbox"/><br>5                                                                    | <input type="checkbox"/><br>4 | <input type="checkbox"/><br>3 | <input type="checkbox"/><br>2 | <input type="checkbox"/><br>1 |

|     |                                                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 8.  | Have you assessed any societal, health, safety, legal and cultural issues in your project? (PO6)                     |  5 |  4 |  3 |  2 |  1 |
| 9.  | Have you presented your project confidently and prepared effective reports? (PO10)                                   |  5 |  4 |  3 |  2 |  1 |
| 10. | Have you completed the project within a time frame and estimated budget? (PO11)                                      |  5 |  4 |  3 |  2 |  1 |
| 11. | Is your project leads you to recognize the need for lifelong learning in the context of technological change? (PO12) |  5 |  4 |  3 |  2 |  1 |
| 12. | Have you conducted the experiments/ analysis to provide valid conclusion? (PO4,Pso1)                                 |  5 |  4 |  3 |  2 |  1 |

**Signature of the Student:**

**Date:**